

CHAPTER 19 · REDOX REACTIONS — COMPLETE REFERENCE SHEET

+n positive −n negative 0 elemental/zero varies variable oxidation state ★ = high priority for test

Species / Ion	Formula	Ox #	Notes
GOLDEN RULES (always apply)			
Pure elements	H ₂ , O ₂ , Fe...	0	Elemental state = always 0
Fluorine	HF, NaF, CF ₄	−1	Always −1; most electronegative
Oxygen (most cmpds)	H ₂ O, CO ₂	−2	Default for O
Oxygen (peroxides)	H ₂ O ₂ , Na ₂ O ₂	−1	Peroxide O–O bond
Oxygen (OF₂)	OF ₂	+2	F more electroneg. than O
H (with nonmetals)	HCl, H ₂ O	+1	Default for H
H (with metals)	NaH, CaH ₂	−1	Metal hydrides
GROUP 1 & 2 METALS (fixed)			
Li, Na, K, Rb, Cs	NaCl, KNO ₃ ...	+1	Always +1 in compounds
Mg, Ca, Ba, Sr, Be	MgO, CaCO ₃ ...	+2	Always +2 in compounds
Aluminum	Al ³⁺ , Al ₂ O ₃	+3	Always +3 in compounds
Zinc	Zn ²⁺ , ZnO	+2	Always +2 in compounds
Silver	Ag ⁺ , AgNO ₃	+1	Almost always +1
NITROGEN (N)			
Ammonia	NH ₃	−3	N fully reduced
Dinitrogen monoxide	N ₂ O	+1	Laughing gas
Nitric oxide	NO	+2	Radical species
Nitrogen dioxide	NO ₂	+4	Brown toxic gas
Nitrite ion	NO ₂ [−]	+3	N in nitrite
Nitrate ion ★	NO ₃ [−]	+5	N in nitrate — max common
SULFUR (S)			
Hydrogen sulfide	H ₂ S	−2	S fully reduced
Sulfur dioxide	SO ₂	+4	Acid rain precursor
Sulfur trioxide	SO ₃	+6	Forms H ₂ SO ₄ with water
Sulfite ion	SO ₃ ^{2−}	+4	S in sulfite
Sulfate ion ★	SO ₄ ^{2−}	+6	S in sulfate — very common
Thiosulfate ion	S ₂ O ₃ ^{2−}	+2	Used in redox titrations

Species / Ion	Formula	Ox #	Notes
HALOGENS: Cl, Br, I (variable!)			
Chloride (binary)	NaCl, HCl	−1	Default when bonded to metal/H
Hypochlorite	ClO [−]	+1	Bleach active ion
Chlorite	ClO ₂ [−]	+3	
Chlorate	ClO ₃ [−]	+5	
Perchlorate ★	ClO ₄ [−]	+7	Max Cl state
Iodate	IO ₃ [−]	+5	Common in titrations
CARBON (C)			
Methane	CH ₄	−4	C fully reduced
Carbon monoxide	CO	+2	Toxic gas
Carbon dioxide ★	CO ₂	+4	Most common C oxide
Carbonate ion ★	CO ₃ ^{2−}	+4	C in carbonate
PHOSPHORUS (P)			
Phosphate ion ★	PO ₄ ^{3−}	+5	Most important P compound
Phosphoric acid	H ₃ PO ₄	+5	P in phosphoric acid
TRANSITION METALS (variable — memorize!)			
Iron(II) / ferrous	Fe ²⁺ , FeO, FeCl ₂	+2	Ferrous — lighter salts
Iron(III) / ferric ★	Fe ³⁺ , Fe ₂ O ₃	+3	Rust is Fe ₂ O ₃
Copper(I) / cuprous	Cu ⁺ , Cu ₂ O	+1	Less common
Copper(II) / cupric ★	Cu ²⁺ , CuSO ₄	+2	Most common Cu state
Manganese(II)	Mn ²⁺ , MnO	+2	Most stable aqueous form
Manganese(IV)	MnO ₂	+4	Common oxidant
Permanganate ★	MnO ₄ [−]	+7	KMnO ₄ — strong oxidizer
Chromium(III)	Cr ³⁺ , Cr ₂ O ₃	+3	Most stable Cr state
Chromate	CrO ₄ ^{2−}	+6	Yellow, basic solution
Dichromate ★	Cr ₂ O ₇ ^{2−}	+6	Orange, acidic solution
Lead(II)	Pb ²⁺ , PbO	+2	Most common Pb state
Lead(IV)	PbO ₂	+4	Car battery cathode
Mercury(I)	Hg ₂ ²⁺	+1	Exists as Hg ₂ ²⁺ dimer
Mercury(II)	Hg ²⁺ , HgCl ₂	+2	Highly toxic
KEY POLYATOMIC IONS			
Ammonium ★	NH ₄ ⁺	+1	N=−3, H=+1; ion=+1
Hydroxide ★	OH [−]	−1	O=−2, H=+1; ion=−1
Peroxide	O ₂ ^{2−}	−1	Each O = −1
Cyanide	CN [−]	−1	C=+2, N=−3
Acetate	CH ₃ COO [−]	−1	Ion charge = −1

CHAPTER 19 · REDOX CONCEPTS, BALANCING & ELECTROCHEMISTRY

Species / Concept	Formula / Example	Ox #	Notes / Details
CORE DEFINITIONS			
Redox reaction			Transfer of electrons between substances; ox numbers change
Oxidation			Atom LOSES electrons → ox number becomes MORE positive (OIL)
Reduction			Atom GAINS electrons → ox number becomes LESS positive (RIG)
Reducing agent			Causes reduction of another — itself gets OXIDIZED
Oxidizing agent			Causes oxidation of another — itself gets REDUCED
ELECTROCHEMICAL CELLS			
Electrolytic cell			Uses ELECTRICITY to FORCE nonspontaneous redox reaction
Voltaic cell			SPONTANEOUS redox reaction PRODUCES electricity (batteries)
Cathode			Reduction occurs; (+) in voltaic, (–) in electrolytic
Anode			Oxidation occurs; (–) in voltaic, (+) in electrolytic
Salt bridge			Allows ion flow between half-cells; prevents mixing
Electrolysis			Forcing nonspontaneous redox with electrical current
Electroplating			Object to plate = cathode; plating metal = anode
BALANCING REDOX: 5-STEP METHOD			
Step 1			Assign oxidation numbers to all atoms
Step 2			Identify which atoms are oxidized (increase) and reduced (decrease)
Step 3			Draw lines between ox/red atoms; note net change in ox number
Step 4			Multiply pairs by coefficients so total Δox = total Δred
Step 5			Use inspection to finish balancing atoms and charge

Species / Concept	Formula / Example	Ox #	Notes / Details
PPT EXAMPLES TO KNOW			
Ex 19-1a (NOT redox)	$\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$		No ox numbers change — NOT a redox reaction
Ex 19-1b (redox)	$2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$		Cl: +5 → –1; O: –2 → 0 — YES redox
Ex 19-2	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$		Na oxidized (0 → +1); H reduced (+1 → 0)
Ex 19-4 balanced	$3\text{HCl} + 2\text{HNO}_3 \rightarrow 3\text{HOCl} + 2\text{NO} + \text{H}_2\text{O}$		Cl: –1 → +1 (ox); N: +5 → +2 (red)
Ex 19-5 half-rxns	$3\text{Ca} \rightarrow 3\text{Ca}^{2+} + 6\text{e}^-$		Anode (oxidation)
	$2\text{Fe}^{3+} + 6\text{e}^- \rightarrow 2\text{Fe}$		Cathode (reduction)
Rusting	$4\text{Fe}(\text{OH})_2 + \text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 4\text{H}_2\text{O}$		Fe: +2 → +3 oxidized; O ₂ reduced
Voltaic (Zn/Cu)	$\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$		~1.1 V; Zn = anode; Cu = cathode
QUICK RULES SUMMARY			
★ = high-priority	Know these cold		Fe ₂ O ₃ , CuSO ₄ , KMnO ₄ , NO ₃ [–] , CO ₃ ^{2–} , SO ₄ ^{2–} , ClO ₄ [–] , PO ₄ ^{3–} , OH [–] , NH ₄ ⁺
Ox # of ion			Sum of all ox numbers in polyatomic ion = ion's overall charge
Neutral compound			Sum of all ox numbers = 0
Redox check			If ANY ox number changes between reactants and products → it IS redox
OIL RIG mnemonic			Oxidation Is Loss of electrons — Reduction Is Gain of electrons

OIL RIG: Oxidation Is Loss · Reduction Is Gain | Cathode = Reduction = C·R | Anode = Oxidation = A·O | Oxidation numbers in neutral compound always sum to 0